

SSC CGL Study Notes

Complete Topic-Wise Concepts, Shortcuts & PYQs

MULTI-SUBJECT COMPILATION

Quantitative Aptitude: Number System, Divisibility, LCM & HCF

Reasoning: Coding-Decoding

English Language: Subject-Verb Agreement & Tenses

General Awareness: Ancient India (Harappa, Maurya, Gupta)

Table of Contents

Part 1 — Quantitative Aptitude: Number System, Divisibility, LCM & HCF

- 1.1 Number System Basics
- 1.2 Divisibility Rules
- 1.3 LCM & HCF — Concepts & Formulas
- 1.4 SSC CGL PYQs — Number System, LCM & HCF

Part 2 — Reasoning: Coding-Decoding

- 2.1 Alphabet Positions (The Foundation)
- 2.2 Types of Coding-Decoding (with Examples)
- 2.3 Shortcuts for SSC CGL Coding-Decoding
- 2.4 SSC CGL PYQs — Coding-Decoding
- 2.5 Practice Questions

Part 3 — English Language: Subject-Verb Agreement & Tenses

- 3.1 Subject-Verb Agreement — 15 Golden Rules
- 3.2 SSC CGL PYQs — Subject-Verb Agreement
- 3.3 Tenses — Complete Notes
- 3.4 SSC CGL PYQs — Tenses
- 3.5 Quick Revision Chart & Practice Questions

Part 4 — General Awareness: Ancient India

- 4.1 Harappan Civilization (Indus Valley)
- 4.2 Maurya Empire
- 4.3 Gupta Empire
- 4.4 Quick Revision Chart — All 3 Dynasties
- 4.5 Practice Questions

Part 1: Number System, Divisibility, LCM & HCF

This is a foundational topic where questions are often conceptual and can be solved with shortcuts.

1.1 Number System Basics

Types of Numbers

- **Natural Numbers (N):** 1, 2, 3, 4...
- **Whole Numbers (W):** 0, 1, 2, 3...
- **Integers (Z):** ..., -3, -2, -1, 0, 1, 2, 3...
- **Prime Numbers:** Numbers greater than 1, divisible only by 1 and themselves (e.g., 2, 3, 5, 7, 11...)
- **Composite Numbers:** Numbers with more than two factors (e.g., 4, 6, 8, 9...)
- **Co-Prime Numbers:** Two numbers with HCF = 1 (e.g., 8 and 15)

Key Facts

- 1 is neither prime nor composite.
- 2 is the only even prime number.

1.2 Divisibility Rules (Must-Know for Quick Solutions)

Divisor	Rule	Example
2	Last digit is even (0,2,4,6,8)	128 → Divisible
3	Sum of digits is divisible by 3	231 (2+3+1=6) → Divisible
4	Last two digits are divisible by 4	1124 (24 ÷ 4 = 6) → Divisible
5	Last digit is 0 or 5	145 → Divisible
6	Divisible by BOTH 2 and 3	234 → Divisible
7	Double the last digit, subtract it from the rest. Result should be 0 or a multiple of 7	343: 34 - (2×3) = 28, 28÷7=4 → Divisible
8	Last three digits are divisible by 8	5112 (112 ÷ 8 = 14) → Divisible
9	Sum of digits is divisible by 9	729 (7+2+9=18) → Divisible
10	Last digit is 0	230 → Divisible
11	Difference between (sum at odd places) and (sum at even places) is 0 or a multiple of 11	121: (1+1) - 2 = 0 → Divisible
12	Divisible by BOTH 3 and 4	144 (sum=9 & last two=44) → Divisible

1.3 LCM & HCF (Concepts & Formulas)

- **HCF (Highest Common Factor):** The greatest number that divides two or more numbers completely.
- **LCM (Least Common Multiple):** The smallest number that is exactly divisible by two or more numbers.

Key Formula

LCM × HCF = Product of the two numbers (valid for two numbers)

Important Rules for Remainder Problems

- **LCM + Remainder:** Smallest number divisible by x, y, z leaving remainder R → $\text{Answer} = \text{LCM}(x, y, z) + R$
- **LCM – Common Difference:** Smallest number that when divided by x, y, z leaves remainders a, b, c, where $(x-a) = (y-b) = (z-c) = K$ → $\text{Answer} = \text{LCM}(x, y, z) - K$
- **HCF for Same Remainder:** Greatest number that divides x and y leaving the same remainder → $\text{Answer} = \text{HCF}(x-y, y-z)$

1.4 SSC CGL PYQs on Number System, LCM & HCF

Type 1: Number System & Divisibility

Q1. If the seven-digit number 8439x53 is divisible by 99, then the value of x is: (SSC CGL 2018)

(a) 9 (b) 4 (c) 3 (d) 6

Answer: (b) 4

A number is divisible by 99 if it is divisible by both 9 and 11. Applying the divisibility rules for 9 and 11 and solving the congruences for x gives the common solution $x = 4$.

Q2. Which among the following numbers is exactly divisible by 11, 13 and 7? (SSC CHSL 2018)

(a) 259237 (b) 259248 (c) 259270 (d) 259259

Answer: (d) 259259

LCM of 11, 13, and 7 = $11 \times 13 \times 7 = 1001$. The number must be a multiple of 1001. $259259 \div 1001 = 259$ (exact), so option (d) is correct.

Q3. Find the greatest number of five digits which when divided by 3, 5, 8, 12 leaves 2 as a remainder. (SSC CGL Prelim 2002)

(a) 99962 (b) 99960 (c) 99958 (d) 99999

Answer: (a) 99962

LCM of 3, 5, 8, 12 = 120. Greatest 5-digit number = 99999. $99999 \div 120 = 833$ remainder 39, so the greatest 5-digit multiple of 120 is $99999 - 39 = 99960$. Adding the remainder 2 gives $99960 + 2 = 99962$.

Type 2: LCM & HCF — Remainder Problems

Q4. Find the greatest number of four digits which when divided by 3, 5, 7, 9 leaves remainders 1, 3, 5, 7 respectively.

(SSC CGL 2012)

(a) 9765 (b) 9766 (c) 9764 (d) 9763

Answer: (d) 9763

LCM – Common Difference Rule: Each divisor minus its remainder gives the same value: $(3-1)=(5-3)=(7-5)=(9-7)=2$. LCM of (3,5,7,9) = 315. The greatest 4-digit multiple of 315 is 9765. Applying LCM – K: $9765 - 2 = 9763$.

Q5. Find the greatest number which divides 108, 124 and 156, leaving the same remainder. (SSC CGL 2021)

(a) 16 (b) 18 (c) 12 (d) 24

Answer: (a) 16

HCF for Same Remainder Rule: Differences: $124-108=16$, $156-124=32$, $156-108=48$. Required number = $\text{HCF}(16, 32, 48) = 16$.

Type 3: LCM & HCF — Direct Application

Q6. If the students of a class can be grouped exactly into 6 or 8 or 10, then the minimum number of students in the class must be: (SSC CGL Prelim 2004)

(a) 240 (b) 180 (c) 120 (d) 60

Answer: (c) 120

The minimum number of students is the LCM of 6, 8, and 10 = 120.

Q7. A and B are two prime numbers such that $A > B$ and their LCM is 209. The value of $A^2 - B$ is: (SSC CGL 2021)

(a) 350 (b) 352 (c) 354 (d) 356

Answer: (a) 350

For two prime numbers, LCM = their product, so $A \times B = 209$. Factorizing $209 = 11 \times 19$. Since $A > B$, $A = 19$, $B = 11$. $A^2 - B = 361 - 11 = 350$.

Part 2: Reasoning — Coding-Decoding

Coding-Decoding is the process of encrypting a word, phrase, or number in a specific pattern. The candidate must identify the pattern and apply it to find the answer.

2.1 Alphabet Positions (The Foundation)

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

Opposite Letter Pairs (Sum = 27)

A↔Z, B↔Y, C↔X, D↔W, E↔V, F↔U, G↔T, H↔S, I↔R, J↔Q, K↔P, L↔O, M↔N

Memorization Trick: Remember the pair "A-Z" and "M-N" — all other pairs fall in between. Opposite of any letter = 27 – Position of that letter. Example: Opposite of G = 27 – 7 = 20 → T.

2.2 Types of Coding-Decoding (with Examples)

SSC CGL asks questions from six main types.

Type 1: Letter Shifting (Addition/Subtraction)

Each letter is shifted forward (+) or backward (–) by a fixed number of positions.

Example 1 (Direct shift): If "DOCTOR" is coded as "FQEVQT", how is "NURSE" coded?

Logic: every letter shifted +2 (D→F, O→Q, C→E, T→V, O→Q, R→T).

Answer: N→P, U→W, R→T, S→U, E→G → **"PWTUG"**

Example 2 (Alternate shift): If "MAN" is coded as "NZM", how is "TOP" coded?

Logic: M→N (+1), A→Z (–1), N→M (–1), i.e., pattern +1, –1, –1.

Answer: T→U (+1), O→P (+1), P→O (–1) → **"UPO"**

Type 2: Opposite Letter Coding

Each letter is replaced by its opposite letter (A↔Z, B↔Y, etc.)

Example: If "DELHI" is coded as "WVOSR", how is "MUMBAI" coded?

Logic: D→W, E→V, L→O, H→S, I→R are all opposite pairs.

Answer: M→N, U→F, M→N, B→Y, A→Z, I→R → **"NFNYZR"**

Type 3: Operation on Place Values (Sum/Difference/Multiplication)

Letters are converted to numbers and an arithmetic operation is performed.

Example (Sum of place values): If "CAT" is coded as 24 (C=3, A=1, T=20 → 3+1+20=24), how is "DOG" coded?

Answer: D=4, O=15, G=7 → 4+15+7 = **26**

Some SSC questions in this category use less obvious combinations (e.g., sums restricted to certain letter groups, or sum minus a fixed offset). When the straightforward sum doesn't match the given code, check the answer options and work backward to find a consistent rule before finalizing.

Type 4: Substitution (Verbal Coding)

Words are replaced by other words — this requires logical thinking rather than a numeric pattern.

Example: If "PEN" is called "BOOK", "BOOK" is called "TABLE", and "TABLE" is called "CHAIR", what do you write with?

Logic: You normally write with a PEN, but PEN is called BOOK.

Answer: **BOOK**

Type 5: Mixed (Reverse + Shift)

The word is reversed first, and then letters are shifted by a fixed amount. These patterns can look irregular at first glance — the reliable approach is to write out the alphabet series and compare each letter of the word against its code letter systematically, rather than assuming a single shift value applies throughout.

SSC rarely asks heavily ambiguous mixed-coding questions. Mastering Types 1–4 covers the large majority of questions in this section.

Type 6: Matrix/Group Coding (Newer Pattern in SSC)

Letters are coded as numbers using a row-column matrix format.

Example: Given a matrix where Row 1 = A B C D and Row 2 = E F G H, if 'A' is coded as '11' and 'F' is coded as '22', how is 'B' coded?
Logic: B is in Row 1, Column 2.
Answer: **12**

2.3 Shortcuts for SSC CGL Coding-Decoding

Type	Shortcut Trick
Direct Shift	Check the difference between the first letter of the word and its code letter; apply the same shift to all letters.
Opposite Letters	Remember the pairs A-Z, B-Y, C-X...; use 27 – Position to find any opposite instantly.
Sum of Place Values	Convert each letter to its position and add/multiply per the given pattern.
Substitution	Read the question carefully and replace the word in the question with the new name given.
Matrix Coding	Find the row and column of the letter in the given matrix; note a letter can have more than one valid representation.

2.4 SSC CGL PYQs on Coding-Decoding

Type 1: Letter Shifting

Q1. In a certain code language, "MADRAS" is written as "NBESBT". How is "DELHI" written in that code? (SSC CGL 2021)
(a) EFMIJ (b) EFMJH (c) EFMJI (d) EFMIH
Answer: (a) EFMIJ
Each letter is shifted by +1 (M→N, A→B, D→E, R→S, A→B, S→T). Applying +1 to D-E-L-H-I gives E-F-M-I-J.

Q2. In a certain code language, "DANGER" is coded as "FCZGDP". How is "BOTTLE" coded in that code? (SSC CHSL 2021)
(a) DQVVNG (b) DQVVNF (c) DQVVMG (d) DRVVNG
Answer: (a) DQVVNG
Checking option (a) against a uniform +2 shift confirms consistency: B→D, O→Q, T→V, T→V, L→N, E→G — every letter shifted by +2.

Q3. In a certain code language, "RESULT" is coded as "SFTVMU". How is "EXAM" coded in that code? (SSC CGL 2020)
(a) FYBN (b) FYBM (c) FYAN (d) FZBN
Answer: (a) FYBN
Each letter is shifted by +1 (R→S, E→F, S→T, U→V, L→M, T→U). Applying +1 to E-X-A-M gives F-Y-B-N.

Type 2: Opposite Letter Coding

Q4. In a certain code language, "PENCIL" is written as "KVMXRO". How is "BOTTLE" written in that code? (SSC CGL 2019)
(a) YLGGOV (b) YLGGVO (c) YLGGOA (d) YLGGOT
Answer: (a) YLGGOV
Each letter is replaced by its opposite (P→K, E→V, N→M, C→X, I→R, L→O). Applying the same rule to B-O-T-T-L-E gives Y-L-G-G-O-V.

Q5. In a certain code language, "GATE" is coded as "TZGV". How is "ROAD" coded in that code? (SSC CPO 2020)
(a) ILZW (b) ILZW (c) ILZV (d) ILZY
Answer: (a) ILZW
Each letter is replaced by its opposite (G→T, A→Z, T→G, E→V). Applying the same rule to R-O-A-D gives I-L-Z-W.

Type 3: Operation on Place Values

Q6. If "PEN" is coded as 35 and "COPY" is coded as 49, then how is "TABLE" coded? (SSC CGL 2022)
(a) 44 (b) 46 (c) 48 (d) 50
Commonly cited answer: (c) 48
PEN's sum of place values (P=16, E=5, N=14) is 16+5+14 = 35, matching its code directly. COPY's raw sum (C=3, O=15, P=16, Y=25 → 59) does not match its code of 49, which indicates an additional adjustment rule for longer words. This question is known to have some ambiguity across sources on the exact adjustment; if it appears in practice, confirm the base logic (sum of place values) from the PEN example and use elimination against the given options.

Type 4: Substitution (Verbal Coding)

Q7. If "ROSE" is called "LILY", "LILY" is called "LOTUS", "LOTUS" is called "JASMINE", and "JASMINE" is called "SUNFLOWER", then which flower is white in colour? (SSC CGL 2020)

(a) ROSE (b) LILY (c) LOTUS (d) JASMINE

Answer: (c) LOTUS

The flower that is actually white is LILY. But per the code, LILY is called LOTUS — so the answer given in the code language is LOTUS.

Type 5: Matrix Coding

Q8. Using a two-matrix coding scheme (Matrix-I rows/columns numbered 0-4 for letters A-Y, Matrix-II rows/columns numbered 5-9 for letters Z, A-X), find the set of numbers for the word "YOUTH". (SSC CGL 2021)

(a) 44, 65, 56, 20, 87 (b) 44, 56, 65, 20, 87 (c) 44, 65, 56, 20, 78 (d) 44, 56, 65, 20, 78

This is a genuinely tricky matrix-coding question because each letter can have more than one valid row-column representation across the two matrices. Y is consistently 44 (Matrix-I, Row 4, Col 4) in all options, which is a useful anchor. For matrix-coding questions generally: first fix the unambiguous letters using the matrix layout, then use those as anchors to test each option systematically rather than trying to derive every letter's code from scratch. Given the ambiguity in the underlying source for this particular item, treat it as a format-familiarization example rather than a definitive PYQ answer key.

2.5 Practice Questions

1. If "MOBILE" is coded as "FNFYQZ", then how is "PHONE" coded?
2. If A=26, B=25, C=24... then what is the value of "SUM"?
3. If "RED" is coded as "QDC", then how is "BLUE" coded?
4. In a certain code, "TIGER" is written as "UJIFS". How is "LION" written?
5. If "DELHI" is coded as "CCKGH", then find the code for "MUMBAI".

Answers:

1. QGJLY (alternating −1 and +1 shift)
2. S=8, U=6, M=14 → 8+6+14 = 28
3. AKTD (−1 shift on each letter)
4. MJPO (+1 shift on each letter)
5. LTLAZH (mixed shift; work through letter-by-letter differences as shown in the alphabet table)

Part 3: English Language — Subject-Verb Agreement & Tenses

3.1 Subject-Verb Agreement (Concord) — 15 Golden Rules

The verb in a sentence must agree in number and person with its subject: Singular Subject → Singular Verb; Plural Subject → Plural Verb.

Rule 1: Singular Subject = Singular Verb | Plural Subject = Plural Verb

✓ He runs fast. ✓ They run fast.

Rule 2: Two singular subjects joined by "and" = Plural verb

✓ Ram and Shyam are going to market. ✓ The teacher and the student have come.

Exception: If they form a single unit/idea, use a singular verb.

✓ Bread and butter is my favourite breakfast. ✓ Slow and steady wins the race.

Rule 3: "as well as", "with", "together with", "along with", "in addition to", "including", "besides" → Verb agrees with the FIRST subject

✓ The teacher, along with the students, is going on a picnic.

✓ The captain, together with the players, was felicitated.

Rule 4: "Either...or", "Neither...nor", "Not only...but also" → Verb agrees with the NEAREST subject

✓ Neither the boys nor their teacher is present. (nearest = teacher → singular)

✓ Neither the teacher nor the boys are present. (nearest = boys → plural)

Rule 5: Indefinite pronouns (Everyone, Someone, No one, Anyone, Each, Every, Either, Neither, Much, Little) = Always Singular

✓ Everyone has his own opinion. ✓ Each of the students has submitted the assignment.

Rule 6: Collective nouns (Team, Jury, Committee, Audience, Flock, Family, Staff) — Singular as a unit, Plural as individuals

✓ The team is playing well. (unit) ✓ The team are fighting among themselves. (individuals)

Rule 7: "The number of" = Singular | "A number of" = Plural

✓ The number of cases is increasing. ✓ A number of people are waiting outside.

Rule 8: Uncountable/abstract nouns (Mathematics, Physics, Economics, News, Information) = Always Singular

✓ Mathematics is a difficult subject. ✓ The news is shocking.

Scissors, Spectacles, Trousers, Jeans, Pants, Shorts always take a **plural** verb (e.g., "These scissors are sharp"), even though they don't fit the "uncountable = singular" pattern above.

Rule 9: "Many a" = Singular

✓ Many a man has tried to climb the mountain.

Rule 10: Distances, periods of time, amounts of money, weights, measures = Singular (as a whole)

✓ Ten kilometres is a long walk. ✓ Five years is a long time.

Rule 11: Relative pronouns (Who, Which, That) → Verb agrees with the antecedent

✓ I am the person who is responsible. ✓ I am one of those persons who are responsible.

Rule 12: Nouns plural in form but singular in meaning

✓ The United States is a powerful country. ✓ The Arabian Nights is an interesting book.

Rule 13: "None" = Singular or Plural, depending on context

✓ None is so blind as those who will not see. ✓ None of the students have submitted the assignment.

Rule 14: "There" + Verb → Verb agrees with the subject that follows

✓ There is a book on the table. ✓ There are many books on the table.

Rule 15: Gerund/Infinitive as subject = Singular

✓ Walking is a good exercise. ✓ To err is human.

3.2 SSC CGL PYQs — Subject-Verb Agreement

Q1. Neither the students nor the teacher ____ present in the class. (SSC CGL 2021)

(a) is (b) are (c) were (d) have

Answer: (a) is

Rule 4: verb agrees with the nearest subject — "teacher" (singular).

Q2. The team, along with the coach, ____ arrived at the stadium. (SSC CHSL 2020)

(a) has (b) have (c) are (d) were

Answer: (a) has

Rule 3: verb agrees with the first subject — "team" (singular).

Q3. A number of students ____ appeared for the examination. (SSC CGL 2019)

(a) has (b) have (c) is (d) was

Answer: (b) have

Rule 7: "A number of" = Plural.

Q4. The number of applicants ____ increased significantly. (SSC CPO 2020)

(a) have (b) are (c) is (d) were

Answer: (c) is

Rule 7: "The number of" = Singular.

Q5. Each of the boys ____ given a prize. (SSC CGL 2022)

(a) were (b) are (c) was (d) have

Answer: (c) was

Rule 5: "Each of" = Singular.

Q6. The jury ____ divided in their opinions. (SSC CHSL 2021)

(a) is (b) was (c) were (d) has

Answer: (c) were

Rule 6: collective noun acting as individuals (divided opinions) → plural verb.

Q7. Many a man ____ tried to climb Mount Everest. (SSC CGL 2020)

(a) have (b) were (c) has (d) are

Answer: (c) has

Rule 9: "Many a" = Singular.

Q8. Ten kilometres ____ not a long distance. (SSC CPO 2019)

(a) are (b) were (c) is (d) have

Answer: (c) is

Rule 10: distance considered as a whole → singular verb.

Q9. The news ____ very disturbing. (SSC CGL 2021)

(a) are (b) were (c) has (d) is

Answer: (d) is

Rule 8: "News" is uncountable → always singular.

Q10. I am one of those teachers who ____ dedicated to their work. (SSC CHSL 2022)

(a) is (b) am (c) are (d) was

Answer: (c) are

Rule 11: "who" refers to "teachers" (plural) → plural verb.

3.3 Tenses — Complete Notes

Tenses indicate the time of an action: Present, Past, and Future — each with 4 sub-types: Simple, Continuous, Perfect, and Perfect Continuous.

1. Present Tense

Type	Structure	Example
Simple Present	Subject + V1 (+s/es)	He reads daily.
Present Continuous	Subject + is/am/are + V1+ing	He is reading now.
Present Perfect	Subject + has/have + V3	He has read the book.
Present Perfect Continuous	Subject + has/have + been + V1+ing + since/for	He has been reading for 2 hours.

Key Rules

- **Simple Present:** habitual actions, universal truths, general facts, scheduled events.
- **Present Perfect:** action started in the past and continues to the present; action just completed; past action with present relevance. Keywords: just, yet, already, ever, never, since, for, recently, so far.
- **Present Continuous:** action happening now, temporary actions, near-future plans.

2. Past Tense

Type	Structure	Example
Simple Past	Subject + V2	He read the book.
Past Continuous	Subject + was/were + V1+ing	He was reading when I called.
Past Perfect	Subject + had + V3	He had read the book before I arrived.
Past Perfect Continuous	Subject + had + been + V1+ing + since/for	He had been reading for 2 hours when I arrived.

Key Rules (Most Important for SSC)

- **Simple Past:** completed action in the past; past habits.
- **Past Perfect:** used when two past actions occur — the earlier action takes Past Perfect, the later takes Simple Past. Keywords: before, after, already, just, till then.
- **Past Continuous:** action in progress at a specific past time; interrupted actions.

3. Future Tense

Type	Structure	Example
Simple Future	Subject + will/shall + V1	He will read the book.
Future Continuous	Subject + will be + V1+ing	He will be reading at 8 PM.
Future Perfect	Subject + will have + V3	He will have read the book by 8 PM.
Future Perfect Continuous	Subject + will have been + V1+ing + since/for	He will have been reading for 2 hours by 8 PM.

Key Rules (SSC Special)

- **Never use "will"/"shall" in the "if" clause:** ✓ If you come, I will go. (not "If you will come...")
- **Use Present Tense in time clauses** (when, as soon as, until, after, before): ✓ I will wait until he comes.
- **"It's time" / "It is high time"** → always followed by Past Tense (V2): ✓ It is high time you went home.
- **"Lest"** → always followed by "should": ✓ Walk slowly lest you should fall.

3.4 SSC CGL PYQs — Tenses

Q1. If you ____ hard, you will pass the exam. (SSC CGL 2021)

(a) work (b) will work (c) worked (d) have worked

Answer: (a) work

Conditional "if" clause never takes "will" — use Present Tense.

Q2. I ____ in this city since 2015. (SSC CHSL 2020)

(a) live (b) have lived (c) lived (d) was living

Answer: (b) have lived

"Since 2015" — action started in the past and continues → Present Perfect.

Q3. The train ____ before we reached the station. (SSC CGL 2019)

(a) left (b) had left (c) has left (d) was leaving

Answer: (b) had left

Earlier of two past actions → Past Perfect.

Q4. She ____ her homework already. (SSC CPO 2020)

(a) has finished (b) had finished (c) finished (d) will finish

Answer: (a) has finished

"Already" is used with Present Perfect.

Q5. I ____ never ____ such a beautiful place. (SSC CGL 2022)

(a) have, seen (b) had, seen (c) did, see (d) was, seeing

Answer: (a) have, seen

"Never" is used with Present Perfect.

Q6. He ____ to the gym every morning. (SSC CHSL 2021)

(a) go (b) goes (c) went (d) will go

Answer: (b) goes

"Every morning" → habitual action → Simple Present.

Q7. When I reached the hall, the movie ____. (SSC CGL 2020)

(a) has started (b) had started (c) started (d) will start

Answer: (b) had started

Earlier of two past actions → Past Perfect.

Q8. I ____ my lunch when the phone rang. (SSC CPO 2019)

(a) ate (b) was eating (c) have eaten (d) had eaten

Answer: (b) was eating

Longer action in progress interrupted by a shorter one → Past Continuous.

Q9. It is high time we ____ the decision. (SSC CGL 2021)

(a) take (b) took (c) have taken (d) will take

Answer: (b) took

"It is high time" is always followed by Past Tense.

Q10. Walk carefully lest you ____ fall. (SSC CHSL 2022)

(a) shall (b) will (c) should (d) would

Answer: (c) should

"Lest" is always followed by "should".

3.5 Quick Revision Chart & Practice Questions

Tense	Structure	Keywords	SSC Trick
Simple Present	V1 (+s/es)	Usually, often, daily, every day	Habitual actions
Present Continuous	is/am/are + V1+ing	Now, at present, Look!, Listen!	Action happening now
Present Perfect	has/have + V3	Just, already, yet, ever, never, since, for	Past action, present effect
Simple Past	V2	Yesterday, last week, ago	Completed past action
Past Continuous	was/were + V1+ing	While, when (interruption)	Action in progress in past
Past Perfect	had + V3	Before, after, already (past)	Earlier of two past actions
Simple Future	will/shall + V1	Tomorrow, next week	Future prediction
Future Perfect	will have + V3	By + time	Completed by a future time

Practice Questions

Subject-Verb Agreement:

1. Neither the manager nor his assistants ____ (was/were) present.

2. The team, along with the coach, ____ (is/are) in the dressing room.
3. A number of people ____ (has/have) gathered outside.
4. The United States ____ (is/are) a developed country.
5. Each of the soldiers ____ (was/were) given a medal.

Tenses:

1. She ____ (live) here since 2015. (Correct the tense)
2. If he ____ (works/will work) hard, he will succeed.
3. I ____ (finish) my work before he arrived.
4. As soon as the train ____ (arrive), we will board it.
5. It is time we ____ (go) home.

Answers — Subject-Verb Agreement: 1. were (nearest = assistants → plural) 2. is (first subject = team → singular) 3. have (a number of = plural) 4. is (United States = singular) 5. was (each of = singular)

Answers — Tenses: 1. has lived 2. works 3. had finished 4. arrives 5. went

Part 4: General Awareness — Ancient India

4.1 Harappan Civilization (Indus Valley Civilization)

Period: 2500 BC – 1900 BC | **Discovered:** 1921 | **Nature:** Bronze Age, Urban, Highly Advanced Drainage System

Major Sites & Discoverers

Site	Location (Modern)	River	Discoverer	Key Findings
Harappa	Punjab (Pakistan)	Ravi	Daya Ram Sahni (1921)	First site discovered, Granaries, Cemetery
Mohenjo-Daro	Sindh (Pakistan)	Indus	R.D. Banerji (1922)	Great Bath, Great Granary, Dancing Girl, Priest-King, Pashupati Seal
Lothal	Gujarat	Bhogava	S.R. Rao (1954)	Dockyard (world's first), Bead-making factory
Dholavira	Gujarat	Luni	J.P. Joshi (1967)	Water reservoir, Stadium, Only site with 3 parts
Kalibangan	Rajasthan	Ghaggar	A. Ghosh (1953)	Ploughed field, Fire altars
Ropar	Punjab	Sutlej	Y.D. Sharma (1953)	Dog burial, Copper tools
Banawali	Haryana	Ghaggar	R.S. Bisht (1973)	Beads, Barley grains
Suktagendor	Balochistan (Pakistan)	Dasht	A. Stein (1927)	Indus-Persian border trading post
Daimabad	Maharashtra	Pravara	B.P. Bopardikar (1958)	Bronze chariot, Elephant, Bull

Key Features of Harappan Civilization

Feature	Detail
Town Planning	Grid pattern, streets cutting at right angles
Drainage System	Covered, underground, highly advanced
Script	Undeciphered, written right to left (Boustrophedon style)
Religion	Worshipped Pashupati (Proto-Shiva), Mother Goddess, Trees (Pipal), Animals (Bull)
Agriculture	Wheat, Barley, Cotton, Rice (Lothal), Dates, Peas
Animals	Bull (most important), Elephant, Rhinoceros, Tiger, Dog, Cat (no Horse evidence)
Trade	With Mesopotamia (Sumer), Oman, Bahrain
Weights & Measures	Cubical weights, binary system (1, 2, 4, 8, 16, 32, 64...)

Decline of the Harappan Civilization

- Climate change (drying up of the Ghaggar-Hakra river)
- Floods (in Mohenjo-Daro)
- Deforestation / overuse of resources
- Earthquakes
- Aryan invasion (most debated theory)

SSC CGL PYQs — Harappan Civilization

Q1. The Great Bath is found at which Harappan site? (SSC CGL 2021)

(a) Harappa (b) Lothal (c) Mohenjo-Daro (d) Dholavira

Answer: (c) Mohenjo-Daro

Q2. Who discovered the Harappan site of Lothal? (SSC CHSL 2020)

(a) Daya Ram Sahni (b) R.D. Banerji (c) S.R. Rao (d) J.P. Joshi

Answer: (c) S.R. Rao

Q3. Which Harappan site has a man-made dockyard? (SSC CGL 2019)

(a) Harappa (b) Mohenjo-Daro (c) Lothal (d) Kalibangan

Answer: (c) Lothal

Q4. The 'Dancing Girl' statue was found at which site? (SSC CPO 2020)

(a) Dholavira (b) Lothal (c) Mohenjo-Daro (d) Harappa

Answer: (c) Mohenjo-Daro

Q5. Which Harappan site has a water reservoir? (SSC CGL 2022)

(a) Lothal (b) Dholavira (c) Kalibangan (d) Ropar

Answer: (b) Dholavira

Q6. The script of the Indus Valley Civilization is: (SSC CHSL 2021)

(a) Deciphered (b) Undeciphered (c) Brahmi (d) Kharosthi

Answer: (b) Undeciphered

Q7. Which animal was NOT known to the Indus Valley people? (SSC CGL 2020)

(a) Bull (b) Horse (c) Elephant (d) Tiger

Answer: (b) Horse

Q8. The Pashupati Seal found at Mohenjo-Daro depicts which god? (SSC CPO 2019)

(a) Vishnu (b) Shiva (c) Brahma (d) Indra

Answer: (b) Shiva (Proto-Shiva)

4.2 Maurya Empire (322 BC - 185 BC)

Founded by: Chandragupta Maurya | Capital: Pataliputra (modern Patna) | First pan-Indian empire

Important Rulers

Ruler	Reign	Important Facts
Chandragupta Maurya	322-298 BC	Founder; defeated Dhanananda (last Nanda king); helped by Chanakya
Bindusara	298-272 BC	Son of Chandragupta; titled Amitraghata ("Slayer of enemies")
Ashoka the Great	268-232 BC	Greatest Mauryan ruler; Kalinga War (261 BC); embraced Buddhism

Chandragupta Maurya

- Founder of the Maurya Empire (322 BC); defeated Dhanananda with Chanakya's help.
- Chanakya (Kautilya/Vishnugupta) wrote the *Arthashastra* — a treatise on economics, politics, and statecraft.
- Megasthenes, the Greek ambassador, visited his court and wrote *Indica*, describing Mauryan administration, army, and society.
- Later became a Jain monk and died at Shravanabelagola (Karnataka).
- Administration followed the Saptanga Theory (7 parts/limbs of the state).

Ashoka the Great (Most Important for SSC)

- Grandson of Chandragupta Maurya.
- Fought the Kalinga War (261 BC) against the Kalinga kingdom (modern Odisha); the massive bloodshed led him to embrace Buddhism.
- Adopted Dhamma — a policy of non-violence, tolerance, religious freedom, and welfare — and appointed Dhamma Mahamattas to spread it.
- 14 Major Rock Edicts: the 13th describes the Kalinga War and his remorse; the 14th describes the Dhamma policy.
- 7 Pillar Edicts, found at Sarnath, Allahabad, and elsewhere. The Sarnath Pillar's four lions are now India's National Emblem.
- Built stupas (Sanchi, Bharhut) and organized the 3rd Buddhist Council at Pataliputra.
- Mauryan art used polished sandstone (the famed "Mauryan polish").

Mauryan Administration

Post	Name
King	Chakravartin
Chief Minister	Mahamantri
Governor of Provinces	Kumar / Kumara
Finance Minister	Sannidhata
Army Commander	Senapati
Spy Chief	Gopas / Prativedaka

Decline of the Mauryan Empire

- Weak successors after Ashoka
- Heavy military expenditure and financial crisis
- Rebellions in provinces
- Pushyamitra Sunga (Commander-in-chief) assassinated the last Mauryan king, Brihadratha, and founded the Sunga dynasty (185 BC)

SSC CGL PYQs — Mauryan Empire

Q1. Who founded the Mauryan Empire? (SSC CGL 2021)

(a) Ashoka (b) Bindusara (c) Chandragupta Maurya (d) Harsha

Answer: (c) Chandragupta Maurya

Q2. Who wrote the Arthashastra? (SSC CHSL 2020)

(a) Megasthenes (b) Chanakya (c) Ashoka (d) Kautilya

Answer: (b)/(d) Chanakya (Kautilya) — both are correct, as they refer to the same person

Q3. The Kalinga War was fought in which year? (SSC CGL 2019)

(a) 261 BC (b) 261 AD (c) 232 BC (d) 268 BC

Answer: (a) 261 BC

Q4. Which rock edict of Ashoka describes the Kalinga War? (SSC CPO 2020)

(a) 1st (b) 13th (c) 10th (d) 14th

Answer: (b) 13th Rock Edict

Q5. The National Emblem of India is taken from the pillar of: (SSC CGL 2022)

(a) Sanchi (b) Sarnath (c) Bodh Gaya (d) Bharhut

Answer: (b) Sarnath

Q6. Which Greek ambassador visited the court of Chandragupta Maurya? (SSC CHSL 2021)

(a) Megasthenes (b) Fa-Hien (c) Hiuen Tsang (d) Al-Biruni

Answer: (a) Megasthenes

Q7. Who was the last ruler of the Mauryan dynasty? (SSC CGL 2020)

(a) Ashoka (b) Brihadratha (c) Bindusara (d) Chandragupta

Answer: (b) Brihadratha

Q8. Ashoka embraced Buddhism after which war? (SSC CPO 2019)

(a) Kalinga War (b) Battle of Hydaspes (c) Battle of Talikota (d) Battle of Kurukshetra

Answer: (a) Kalinga War

4.3 Gupta Empire (320 AD - 550 AD)

Founded by: Chandragupta I | **Capital:** Pataliputra | Known as the Golden Age of India

Important Rulers

Ruler	Reign	Important Facts
Chandragupta I	319-335 AD	Founder; started the Gupta Era (320 AD); married Kumaradevi (Lichchhavi princess)
Samudragupta	335-375 AD	"Napoleon of India"; great conqueror; also a poet/musician
Chandragupta II (Vikramaditya)	375-415 AD	Greatest Gupta ruler; Golden Age peak; Fa-Hien visited India
Kumaragupta I	415-455 AD	Founded Nalanda University
Skandagupta	455-467 AD	Last great ruler; defended against the Hunas (White Huns)

Samudragupta ("Napoleon of India")

- Conquered many kingdoms across North India (Aryavarta) and South India (Dakshinapatha).
- The Allahabad Pillar Inscription, composed by Harisena, details his conquests.
- A follower of Hinduism (Vishnu); performed the Ashvamedha Yajna (horse sacrifice) to affirm his sovereignty.

- Also known for his skill in music (playing the harp) and poetry.

Chandragupta II (Vikramaditya) — Most Important for SSC

- Greatest Gupta ruler; title Vikramaditya means "Valour like the sun."
- Fa-Hien, a Chinese Buddhist pilgrim, visited India during his reign (399–414 AD) and wrote *Fo-Kwo-Ki* (Record of Buddhist Kingdoms).
- His court held the Navaratnas (nine gems): Kalidasa (poet — *Abhijnanasakuntalam*, *Meghaduta*), Aryabhata (mathematician-astronomer), Varahamihira (astronomer — *Brihat Samhita*), Amarsimha (*Amara Kosha*), Dhanvantari (physician, associated with the origins of Ayurveda), among others.
- Erected the Iron Pillar of Delhi, famous for resisting rust for over 1500 years.
- Conquered western Gujarat (Saurashtra), Malwa, and Ujjain, which became a major cultural and trade centre.

Achievements of the Gupta Period

Aspect	Detail
Administration	Decentralized, feudal system, provincial governors (Kumar)
Economy	Flourishing trade with the Roman Empire, China, South-East Asia
Coinage	Gold coins (Dinars), known for their purity
Science	Aryabhata — concept of zero, value of Pi (3.1416), heliocentric theory
Astronomy	Varahamihira — <i>Brihat Samhita</i>
Literature	Kalidasa — <i>Abhijnanasakuntalam</i> , <i>Meghaduta</i> , <i>Kumarasambhava</i> ; Vishakhadatta — <i>Mudrarakshasa</i> ; <i>Panchatantra</i> by Vishnu Sharma
Medicine	Dhanvantari — associated with the origins of Ayurveda; Sushruta — <i>Sushruta Samhita</i>
Art	Ajanta and Ellora cave paintings, Dashavatara Temple (Deogarh)
Education	Nalanda University (founded by Kumaragupta I)
Metallurgy	Iron Pillar of Delhi — rust-resistant

Decline of the Gupta Empire

- Huna invasions (White Huns) — Toramana, Mihirkula
- Weak successors after Skandagupta
- Financial crisis due to prolonged wars
- Feudal governors gradually asserting independence, leading to disintegration into smaller kingdoms

SSC CGL PYQs — Gupta Empire

Q1. Who founded the Gupta Empire? (SSC CGL 2021)

(a) Samudragupta (b) Chandragupta I (c) Chandragupta II (d) Kumaragupta

Answer: (b) Chandragupta I

Q2. Who is known as the 'Napoleon of India'? (SSC CHSL 2020)

(a) Ashoka (b) Chandragupta II (c) Samudragupta (d) Harsha

Answer: (c) Samudragupta

Q3. The Chinese traveler Fa-Hien visited India during the reign of: (SSC CGL 2019)

(a) Ashoka (b) Chandragupta II (c) Samudragupta (d) Harsha

Answer: (b) Chandragupta II

Q4. Who wrote 'Abhijnanasakuntalam'? (SSC CPO 2020)

(a) Aryabhata (b) Kalidasa (c) Varahamihira (d) Vishakhadatta

Answer: (b) Kalidasa

Q5. The Iron Pillar of Delhi was erected by which ruler? (SSC CGL 2022)

(a) Ashoka (b) Samudragupta (c) Chandragupta II (d) Harsha

Answer: (c) Chandragupta II

Q6. Which university was founded during the Gupta period? (SSC CHSL 2021)

(a) Takshashila (b) Nalanda (c) Vikramshila (d) Vallabhi

Answer: (b) Nalanda

Q7. The 'Navaratnas' (9 gems) were in the court of which ruler? (SSC CGL 2020)

- (a) Ashoka (b) Samudragupta (c) Chandragupta II (d) Harsha
Answer: (c) Chandragupta II

Q8. Who wrote the famous book 'Arthashastra'? (SSC CPO 2019)

- (a) Kautilya (b) Kalidasa (c) Vishakhadatta (d) Banabhatta
Answer: (a) Kautilya (Chanakya)

Q9. Which Gupta ruler is also known as 'Vikramaditya'? (SSC CGL 2021)

- (a) Chandragupta I (b) Samudragupta (c) Chandragupta II (d) Kumaragupta
Answer: (c) Chandragupta II

Q10. Which inscription describes the conquests of Samudragupta? (SSC CHSL 2022)

- (a) Sarnath Inscription (b) Sanchi Inscription (c) Allahabad Pillar Inscription (d) Junagadh Inscription
Answer: (c) Allahabad Pillar Inscription

4.4 Quick Revision Chart — All 3 Dynasties

Feature	Harappa (IVC)	Maurya	Gupta
Period	2500-1900 BC	322-185 BC	320-550 AD
Founder	-	Chandragupta Maurya	Chandragupta I
Greatest Ruler	-	Ashoka	Chandragupta II (Vikramaditya)
Capital	-	Pataliputra	Pataliputra
Religion	Proto-Shiva, Mother Goddess	Buddhism (Ashoka)	Hinduism (Vishnu)
Script	Undeciphered	Brahmi, Kharosthi	Brahmi, Gupta script
Major Site	Mohenjo-Daro (Great Bath)	Sarnath (Pillar)	Ujjain, Nalanda
Foreign Invaders	-	Greeks	Hunas
Art/Literature	Seals, Pottery	Polished stone pillars	Iron Pillar, Ajanta paintings
Scientist	-	-	Aryabhata (Zero, Pi)
Poet	-	-	Kalidasa
Travelers	-	Megasthenes (Greek)	Fa-Hien (Chinese)

4.5 Practice Questions

Harappa:

- The 'Great Bath' is found at which Harappan site?
- Who discovered the Harappan site of Dholavira?
- Which animal was not known to the Indus Valley people?
- What is the script of the Indus Valley called?
- Where is the world's oldest man-made dockyard located?

Maurya:

- Who founded the Mauryan Empire?
- Who wrote the Arthashastra?
- Which war made Ashoka embrace Buddhism?
- Which rock edict of Ashoka describes the Kalinga War?
- What is the National Emblem of India taken from?

Gupta:

- Who founded the Gupta Empire?
- Who is known as the 'Napoleon of India'?
- Which Chinese traveler visited India during Chandragupta II's reign?
- Who wrote 'Abhijnanasakuntalam'?
- Which university was founded during the Gupta period?

Answers — Harappa: Mohenjo-Daro · J.P. Joshi · Horse · Undeciphered · Lothal

Answers — Maurya: Chandragupta Maurya · Chanakya/Kautilya · Kalinga War · 13th Rock Edict · Sarnath Pillar

Answers — Gupta: Chandragupta I · Samudragupta · Fa-Hien · Kalidasa · Nalanda